BORED TUNNELING FOR SINGAPORE METRO

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ABSTRACT: This paper describes the planning, layout, and implementation of the Singapore Metro mass-transit system, emphasizing the criticality of the 22.4 km of bored tunneling required in the central area of the city. The original nine-year construction schedule and the subsequent seven-year accelerated program that was adopted are discussed, as are the extent of the tunneling works and the limited construction period of 3-1/3 years. The contractual arrangements for carrying out the works are outlined. The opportunity afforded by this concentration of soft-ground tunneling activity to compare performance in 38 tunnel drives, all of which excavated almost simultaneously in widely varying geological conditions, is considered unique. Factual details are provided covering the geological conditions along the tunnel drives, the contract packages, the tunnel profile, the tunneling methods, compressed-air equipment and procedures, spoil-handling equipment, temporary and permanent support, and water-sealing techniques. Settlements are reviewed, and tunneling performance for each drive is detailed. Some general conclusions are drawn from the owner's viewpoint to assist in ensuring the timely and efficient completion of tunneled works in a complex transportation project.

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INTRODUCTION

The Singapore Mass Transit Project had a leisurely gestation of some 14 years. The project bore a marked resemblance to many other 20th century projects on paper—from its rather hesitant beginning in 1968 to the authorization of construction in May 1982 by the Singapore government. In this case, however, progress may have been labored because there had been many calls on what was initially a very limited purse—public housing, schools, drainage, water supply, and many other basics had to have prior claim.

Nevertheless, successful industrial and commercial policies and prudent fiscal management meant that the basic needs were fulfilled, and adequate resources were then assembled to provide the infrastructure for an efficient, modern, and highly industrialized state.

The result of the planning was a mass-transit railway of some 67 km and 42 stations (Fig. 1). Operationally the system is divided into two lines, a north-south line and an east-west line, which run concurrently through City Hall and Raffles Place station to provide convenient cross-platform interchange.

The budget cost estimate in 1982 was S$5 billion with the final cost being about same. (Note: S$1 = $0.56 U.S. [approximate].) After allowing for inflation, this represents a total expenditure substantially below the original estimate.

PROGRAM

Construction scheduling was based on a notional average annual expenditure on the railway of 5% of the national development budget. This

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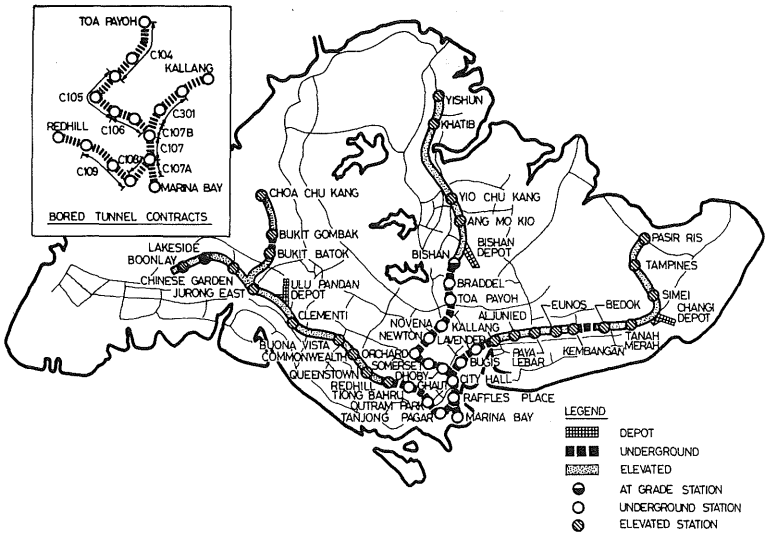


FIG. 1. Singapore Mass Rapid Transit System

resulted in a proposed nine-year construction period, and the program was subdivided into five main phases:

- 1—Yio Chu Kang to Outram Park.
- 1A—Outram Park to Clementi
- 2A—City Hall to Pasir Ris, Raffles Place to Marina Bay.
- 2B—Clementi to Lakeside, Jurong East to Choa Chu Kang, Yio Chu Kang to Yishun.
- 2C—Lakeside to Boon Lay.

This program required a concentration of soft-ground tunneling resources in an Asian city with little previous tunneling experience. However, the government's assessment of the world economy in 1983 proved accurate, and most of the world's soft-ground tunneling contractors were desperately short of work. Consequently the prices obtained for the phase 1 contracts during 1983–84 were so competitive that the government asked the Mass Rapid Transit Corp., Singapore to review and accelerate the whole construction schedule where possible. The corporation concluded that the overall program could be shortened by two years provided that phase 2 contracts could be awarded by 1985. Government subsequently approved this accelerated schedule.

The result was to compress the tunneling program so that 22.4 km of the 5.2-m-inside-diameter (i.d.) bored tunnels in 38 drives were scheduled for completion in 3.5 years including design. While the tunneled length is not remarkable, the large number of drives, the widely varying ground conditions, the fact that almost all tunnels passed through densely developed areas, and the short construction period combined to make the project unique. This is especially true because, at the same time as this bored tunneling activity, 11 km of cut-and-cover tunnels, 41 km of double track viaduct, and 42 stations had to be constructed. Tunneling actually com-

menced on March 1, 1984, with the northbound drive from Shan Road to Toa Payoh. The last breakthrough took place at Bugis Station on the westbound drive from the shaft near City Hall, and this was made on June 18, 1987.

Contractual Arrangements

The tunneling contracts were all of the design-and-construct type. The bored tunneling works, usually combined with the associated station works, were divided into nine packages, each of a size thought likely to attract international interest (values were between S$50,000,000 to S$100,000,000). Tenders were invited from lists of six to eight prequalified contractors drawn up to avoid any one organization obtaining more than two tunneling contracts. This was desirable because any single contractor or joint venture would not become overcommitted or gain a too-dominant position in this critical activity.

The general conditions of contract were based on the *Conditions of Contract for Works of Civil Engineering Construction* (1977) and modified to reflect the design-and-construct approach and conform to the existing construction legislation in Singapore.

The corporation's requirements were detailed in a general design criteria and performance specification, with a particular specification for each individual contract. The one basic requirement was for a 5.2-m-i.d. clear cross-section tunnel correctly aligned after allowing for all design and construction tolerances. To give contractors design freedom, they were required to provide their own materials and workmanship specification, subject to compliance with the corporation's basic criteria.

Contractors were required to submit a cost-loaded critical-path network, and payments were made on completion of the specific activities.

Of the seven major tunneling contracts, three were awarded to Japanese consortia (Tobishima/Takenaka joint venture [JV] [C104], Ohbayashi/Okumura JV [C109], and Taisei/Shimizu/Marubeni JV [C108]), two to Japanese-Singaporean joint ventures (Kajima/Keppel JV [C107] and Nishimatsu/Lum Chang JV [C301]), and two to French-led joint ventures (Borie/Cogefar/Traylor Bros/OCK Construction JV [C105], and Campenon Bernard/Singapore Piling and Civil Engineering JV [C106]).

Geological Conditions along Line of Tunnel Drives

The stratigraphy of Singapore is in three main divisions:

1. Kallang formation—recent deposits of soft clays and fluvial sands.
2. Jurong formation—sedimentary series, comprising mainly varying degrees of weathered sandstone ranging from competent rock through highly fractured/friable hard clay to medium stiff clay.
3. Bukit Timah formation—granite; fresh at depth; completely weathered approaching the ground surface, producing a silty clay.

Tunneling for the Metro passed through all three formations.

The main problem for the tunneling work was the variability in ground conditions along the route and within each tunnel drive. This variability was evident in the different soil types and the widely ranging soil parameters within any one type. For example, the tunnel from Novena to Toa Payoh

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(some 1,200 m long) passed through fresh sandstone, weathered sandstone, marine clay, fluvial sands, estuarine material, completely weathered granite, and fresh granite.

Within each soil type, the parameters of most concern to tunnelers varied considerably; these are represented in Table 1. From a practical point of view, the main soil types encountered are:

1. Marine clays—very soft to soft, varying with depth; sticky and not easy to handle on conveyor belts or with clay spades; require compressed air or ground treatment to form a stable face in an open shield. The preferred way of dealing with the marine clay was simply to push the shield into it in a blind-type approach.
2. Completely decomposed granite—medium stiff to stiff silty clay; relatively low permeability; standard penetration tests (SPTs) range from 5 to greater than 50 at 15 m deep; compressed air preferred with an open shield; boulders of fresh granite encountered in certain areas.
3. Fluvial sands—very variable; range from loose, single-size medium sands in lenses several meters thick to stiff clayey sands with reasonably high cohesion and low permeability; problems of raveling and air loss.
4. Weathered sedimentaries—similar to the completely decomposed granite when completely weathered, except that they are somewhat stiffer and of a lower permeability. Free air tunneling is possible. Beds of boulders were encountered in many areas. These are now thought to be one colluvium deposit rather than different degrees of weathering within the parent material. The boulders can be very strong with an absence of jointing and compressive strengths of between 55 and 185 MN/m^2. The moderately weathered sandstone appeared as a hard clay, but is very friable and gave raveling problems in the tunnel face.

As is normal in the tropics, ground water was virtually at ground level. Because of the risk of inducing wide spread settlement, ground-water lowering was only permitted in very exceptional circumstances, and special precautions were imposed. Tunneling therefore generally had to assume the possibility of full hydrostatic head.

There were no naturally occurring gas problems, but in contract 108, high

TABLE 1. Soil Parameters

Soil type (1)	Undrained strength (Cu) (2)	Standard penetration test—N value (3)	Permeability (4)
M (marine clay)	5–60 kN/m^2	1 to 5	4×10^{-9} to 4×10^{-7} cm/s
F (fluvial sands)—range from pure sand to sandy clay	30–100 kN/m^2 (sandy clay)	1 to 25	4×10^{-8} to 5×10^{-1} cm/s
G4 (completely decomposed granite)	50–200 kN/m^2 at 15 m deep	5 to greater than 50 at 15 m deep	4×10^{-8} to 2×10^{-5} cm/s
S4 (completely weathered sedimentary)	45–300 kN/m^2	10 to greater than 50	5×10^{-6} to 2×10^{-8} cm/s

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temperatures (up to 45°C) and ammonia gas concentrations of 35 ppm were encountered. These were due to the large volume of cement slurry jet grouting and its reaction with the marine clay. Both problems were dealt with by increasing the volume of the tunnel ventilation (Berry et al. 1987).

Tunneling Methods and Face Equipment

The majority of contractors opted for shield machines and the majority of those used were of the Greathead variety fitted with hydraulic backhoe excavators of varying capabilities or roadheaders/boom cutters (Table 2). The relatively short lengths of the drives probably influenced contractors to choose simple, rapidly erected shields.

In summary, Greathead shields fitted with backhoe excavators were used on contracts 104, 105, 106, 107, 107B, 108 (in marine clay lengths), and 301. (Note: Contracts are denoted with the prefix "C.") Greathead shields fitted with roadheaders, or boom cutters were used on C108 (in sandstone lengths) and C109. Blind shields were employed on C104 but proved unsuccessful and had to be modified to semiblind with backhoe excavators. A tunnel-boring machine (TBM) was used by C105. Earth-pressure balance shields (EPBSs) were used on C301.

Marked differences in tunnel construction philosophy between the Japanese and European contractors were evident. Japanese contractors opted for 5.3 or 5.4 m i.d. allowing greater tolerance in shield control; European-led joint ventures chose 5.23 m i.d. In the reuse of shields, apart from C109, all the Japanese contractors provided a new shield machine or a new shield skin for each tunnel drive; even the relatively expensive EPBS machines were scrapped and the skins left in place after only 800 m of drive. The European-led contracts, C105 and C106, originally provided one machine on each contract to construct all four drives. Although good tunneling rates were achieved on both of these contracts, early delays were not recouped sufficiently for the project program, and both contracts had to bring in a second shield.

In a few areas where the ground was competent (compressive strengths up to 50 MPa) or where short lengths of tunnel were involved, shields were not used and a variety of new Austrian tunneling method (NATM)–type solutions were proposed. Excavation equipment varied from the use of a tracked roadheader to minibackhoes and hand excavation in a multidrift method.

The local construction work force is relatively inexperienced in soft-ground tunneling. Consequently the work force in the tunnels was largely international, with Japanese, British, American, Australian, and Singaporean engineers; Korean, American, Hong Kong Chinese, British, and Irish pit bosses and miners; and Malaysian, Korean, Indian, Thai, and Filipino support crews. Care had to be exercised to ensure that effective communication between all the races was possible at all times.

Directional control of shields used lasers and prediction systems (Glennie 1987).

Spoil-Handling Methods and Equipment

Spoil handling (Table 3) within the tunnel drives in all but two instances was carried out with rail systems using battery locomotives and wagons.

Rail of 762-mm gauge and 22 kg/m was fairly standard, and was laid on

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TABLE 2. Shield Details

Contract (1)	Section (2)	Manufacturer (3)	Type (4)	Number of shields (5)	Diameter (external) (m) (6)	Length (including tail skin) (m) (7)	Skin thickness (mm) (8)	Clearance (mm) (9)	Segmental (s) or welded (w) (10)
C104	Shan Road–Toa Payoh	Komatsu	Greathead	2	5.93	5.30	40	25	W
C104	Novena–Newton	Hitachi Zosen	Blind[a]	2	5.93	5.30	40	25	W
C105	Orchard–Newton, Orchard–Somerset (southbound)	Hitachi Zosen	Tunnel-boring machine	1	6.41	6.63	—	NA[b]	S
C105	Orchard–Somerset (northbound)	Grosvenor	Greathead	1	5.85	5.95	35	40	S
C106	Dhoby Ghaut–Somerset, Dhoby Ghaut–City Hall	Grosvenor	Greathead	2	5.85	5.95	35	40	S
C107	Raffles Place–City Hall	Keppel-Mitsui	Greathead	3[d]	5.97	5.39	40	20	W
C108	Tanjong Pagar–Raffles Place	Mitsubishi	Greathead	2	6.03	6.85	40	25	W
C109	Tiong Bahru–Outram, Outram–Tanjong Pagar	Mitsubishi	Greathead	2	5.86	5.41	36	25	W
C107B/C301	City Hall–Bugis	Nishi-matsu–Hiratsuka	Greathead	2	5.92	5.05	40	25	W
C301	Lavender–Bugis	Kawasaki	Earth-pressure balance shield	2	5.93	6.19	40	25	W

[a]Converted to semimechanical.
[b]Ribs expanded against soil at rear of tail skin.
[c]One shield was fitted with five hydraulically operated breasting plates hinged to the hood rams for additional face support.
[d]Four skins.
[e]Originally two face rams, five more added after to improve face support.
[f]With overcutters.
Note: NA = not applicable.

steel joist sleepers at 1-m centers in the segmentally lined tunnels. Rail configuration was usually single track with crossings at appropriate intervals. On some of the shorter lengths such as C107, twin track was laid throughout. With this rail system battery locomotives varied from 6 to 10 metric tons,

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Tail Seals		Shove Rams			Face Rams		Forepolling Rams		Cutter Head			
Number (11)	Type (12)	Number (13)	Thrust (tonne) (14)	Stroke (mm) (15)	Number (16)	Stroke (mm) (17)	Number (18)	Stroke (mm) (19)	Outside diameter (m) (20)	Revolutions per minute (21)	Torque (tonne-m) (22)	Motors (number and power) (23)
1	Natural rubber	24	100	1,150	6	1,000	7	1,000	NA	NA	NA	NA
1	Natural rubber	24	100	1,150	6	850	7	850	NA	NA	NA	NA
None	—	20	125	1,524	NA	—	NA	—	6.41	1.88–4.23	176/394	6 × 150/kw
2	Poly-urethane	18	100	1,300	4	1,000	7	1,000	NA	NA	NA	NA
2	Poly-urethane	18	100	1,300	4	1,000	7[c]	1,000	NA	NA	NA	NA
1	Stainless-steel plates	24	100	1,150	4	800	9	800	NA	NA	NA	NA
2	Wire-brush	24	100	1,150	8	1,000	9	1,000	NA	NA	NA	NA
1	Natural rubber	20	60	1,150	(2)7[c]	1,000	5	1,000	NA	NA	NA	NA
None	—	20	100	1,150	6	1,000	7	1,000	NA	NA	NA	NA
2	Wire-brush	24	125	1,150	NA	—	NA	—	5.91/6.05[f]	0–0.84	Maximum 382	8 × 55 kw

and wagons varied from 3 m^3 to 5 m^3. The exceptions to this were C105, C106, and the NATM section of C104.

On C105, the track was laid directly on the invert of the temporary ribs and laggings lining. Locomotives weighing 25 metric tons and pulling 10-

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TABLE 3. Spoil-Handling Details

Contract (1)	Section (2)	Excavation methods/ equipment type (3)	Excavation Equipment				
			Manufacturer (4)	Capacity (m^3/hr) (5)	Face coverage (6)	Power (7)	Digging force (8)
C104 (shield)	Novena–Newton, Shan Road–Toa Payoh	Shield-mounted backhoe	Komatsu/ Hitachi	35	60%	45 kw	35 tonnes
C104 (NATM)	Shan Road–Novena	Roadheaders with gathering arms and scraper chain conveyor	Mitsui	180	60%	125 kw	Maximum rock strength 600 kg/cm^2
C105	Somerset–Orchard-Newton	Tunnel-boring machine	Hitachi Zosen	220	100%	Six motors at 150 kw each	394 T-M (torque)
C106	City Hall–Dhoby Ghaut–Somerset	Shield-mounted rotary backhoe	Grosvenor	30 (rock)	100%	155 kw	50 tonnes
C107	Raffles Place–City Hall	Shield-mounted backhoe	Mitsui	Backhoe bucket size 0.42 m^3	100% (overcut of 270 mm)	55 kw	32 tonnes
C108	Tanjong Pagar–Raffles Place	Shield-mounted boom cutter/backhoe[a]	Mitsubishi	40	100% (overcut of 100/ 150 mm)	90/75 kw	53 tonnes
C109	Tiong Bahru–Outram–Tanjong Pagar	Shield-mounted roadheader with scraper chain conveyor	Mitsubishi	150	100% (overcut of 200 mm)	75 kw	6.7 tonnes
C301	City Hall–Bugis	Shield-mounted backhoe/rocker shovel RS 95A[b]	Mitsui	RS 95A—0.6 m^3 bucket	60%	37 kw	15 tonnes
C301	Lavender–Bugis	Earth-pressure balance shield	Kawasaki	200	100% (overcut of 120 mm)	Eight motors at 55 kw each	382 T-M (torque)

[a]Boom cutter changed with backhoe part way through drive.
[b]Rocker shovel used initially in westbound drive with open shield later changed to backhoe.

m^3 skips were used. On C106, free-air tunneling was planned by the contractor, and his spoil-removal system from the shield was by using a wheeled Volvo dumper of 12-m^3 capacity driving directly on the precast concrete lining. The NATM section of C104 used a Cat 951 wheeled loader for spoil removal behind the roadheader.

Shaft arrangements for spoil removal were divided equally between those contractors who chose a spoil pit at the base of the shaft with a grab system to surface hoppers and those who chose to lift wagons directly to the surface and empty them into a spoil pit.

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Soil-removal system—face (9)	Spoil-Removal System—Tunnel				Spoil-removal system—shaft (14)	Spoil-storage surface (15)	Material delivery (16)
	Locomotives (10)	Wagons		Rail system (13)			
		Type (11)	Capacity (m^3) (12)				
Belt Conveyors (No. 2)	6 tonnes	Side tipping	4.5	762-mm gauge rail, 22 kg/m	70-m^3 spoil pit at shaft base, 2.5-m^3 grab	No. 2 30-m^3 hopper	7.5T gantry crane
Scraper arms onto chain conveyor and belt conveyor	Cat 951 wheeled loader	Cat 951 wheeled loader	—	None	70-m^3 spoil pit at shaft base, 2.5-m^3 grab	No. 2 30-m^3 hopper	7.5T gantry crane
Belt conveyors (No. 2)	25T diesel (free air) or electric (compressed air)	Box	10	1.0 m gauge	150T link belt crane (fixed) lifting wagons to surface	Surface spoil bin	150T link belt crane
Scraper chain and belt conveyor	Volvo BM861 dumper	Volvo BM861 dumper	12	None	12-m^3 bucket raised on 30T gantry crane	200-m^3 hopper	Same gantry crane; delivery to face by ASTRA truck
Belt conveyors (No. 2)	6 tonnes	Side tipping	3	762-mm gauge rail, 22 kg/m	Muck pit at shaft base; grab (2.5 m^3) to surface	No. 2 30-m^3 hopper	8.7T gantry crane
Belt conveyors (800-mm wide) (No. 2)	10 tonnes	Box	4.5	762-mm gauge rail, 22 kg/m	Wagons lifted to surface by tower crane (8T)	Surface stockpile	8T tower crane
Belt conveyors (850 mm wide) (No. 2) plus backhoe for loading	6 tonnes/8 tonnes	Box	2.5	762-mm gauge rail, 22 kg/m	Wagons raised to surface by 5T gantry crane	65-m^3 hopper	5T gantry crane
Rocker shovel or belt conveyors (No. 2) (750 mm wide)	8 tonnes	Box	4.5/5	762-mm gauge rail, 22 kg/m	Wagons raised to surface by 12T gantry crane	100-m^3 muck bin	6T luffing tower crane
Screw conveyor (780 mm diameter) discharging onto belt conveyor	8 tonnes	Box	4.5/5	762-mm gauge rail, 22 kg/m	Wagons raised to surface by 12T gantry crane	100-m^3 muck pit	6T luffing tower crane

Most working shafts provided a separate gantry or tower crane for materials.

TEMPORARY SUPPORT

About 4,500 m of tunnel was constructed with a temporary lining, followed by a permanent in situ reinforced concrete lining on completion of the drives (Table 4). The most significant section of temporary lining was used on C105 between Somerset to Orchard and Orchard to Newton. In this section, circular steel ribs were erected directly behind the full-face

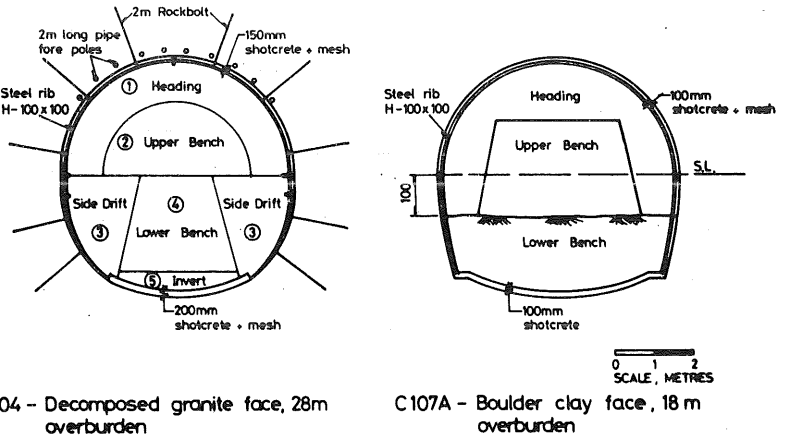
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TABLE 4. Temporary Support Details

Contract (1)	Section (2)	Length (mm) (3)	Ground conditions (4)	Support (5)	Other temporary measures (6)	Excavation method (7)
C104	Shan Road Shaft–Novena	800	Sedimentary rock—fresh to moderately weathered	Shotcrete = 75; mesh = 100 × 100; rockbolts (2–5 m long)	None	Tracked roadheadler, top heading and bench
C104	Shan Road Shaft–Novena	240	Completely decomposed granite	Horseshoe H section arch ribs 125 × 125 at 0.8-m centers, shotcrete = 150; mesh = 100 × 100; rockbolts = 2 m; forepolling bolts = 10 No/m 2-m long	Compressed air	Electric backhoe and hand multidrift
C105	Orchard–Newton and Orchard–Somerset (southbound)	2,680	Completely decomposed granite/marine clay	Circular H section ribs 125 × 125 at 1.2-m centers in decomposed granite and 0.9-m centers in marine clay; ribs expanded onto soil; timber lagging 100 × 100	Compressed air	Tunnel-boring machine (full face)
C105	Orchard–Somerset (northbound)	60	Completely decomposed granite	Horseshoe H section arch ribs 125 × 125 at 1-m spacing; shotcrete = 100/125; mesh = 100 × 100	None	Backhoe—top heading and bench
C107B	City Hall–Bugis and City Hall–C106 interface	460	Boulders in stiff clay	Horseshoe H section arch ribs 125 × 125 at 1.0-m centers; shotcrete = 75; mesh = 100 × 100; steel lagging as necessary	None	Four drives involved—generally minibackhoe and hand in top heading and bench
C107A	Raffles Place–C310 Interface	306	Boulders in stiff clay	Horseshoe H section arch ribs 100 × 100 at 0.5-m centers; shotcrete = 150	Ground treatment (part)	Backhoe—top heading and bench
C301	Working shaft–Lavender	20	Marine clay and sandy clay	Horseshoe H section arch ribs 125 × 125 at 0.8-m centers; shotcrete = 75; mesh = 100 × 100	Lime pile ground treatment	Minibackhoe and hand in multidrift method

Note: All dimensions in millimeters except where shown otherwise.

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C104 - Decomposed granite face, 28m overburden

C107A - Boulder clay face, 18 m overburden

FIG. 2. New Austrian Tunneling Method—Support and Excavation Sequence

shield, and 125-mm-by-100-mm hardwood timber laggings were placed between adjacent ribs to give a fully lagged circular tunnel. Problems of water ingress were encountered, and polythene sheets with silicate grouting were provided on the outside of the laggings.

In other sections, NATM tunneling methods were adopted in ground conditions varying from competent sandstone encountered between Shan Road and Novena on C104 to the lime-pile–treated marine clay between the working shaft and Lavender Station on C301. In the competent rock, the temporary support consisted of shotcrete, mesh, and rock bolts where necessary. The number of bolts was subject to the results of convergence monitoring. The remaining sections of NATM incorporated arch ribs with shotcrete thicknesses from 75 to 150 mm. In all cases, the invert was closed by an additional thickness of shotcrete, generally 200 mm. Details of the temporary support and excavation sequences for C104 and C107A are given in Fig. 2.

COMPRESSED-AIR EQUIPMENT

Compressed-air working was carried out on seven of the contracts (Table 5). The highest pressure was used between City Hall and Bugis and reached 2.35 bars.

The compressed-air equipment provided by the contractors generally had an installed capacity of 5 m^3/min per square meter of face, with standby compressors and generators.

Medical supervision for this examination and treatment of compressed air workers was provided by the Diving Medical Centre of the Singapore Navy. Singapore regulations permit continuous shifts, although the decompression periods at higher pressures reduce the length of time spent in the tunnel (How and Khoo 1987).

Medical locks were provided at each compressed-air work site, and the incident rate of decompression sickness per man decompression was reasonably low, ranging from 0.01% on C105 to 0.34% on C107. Cooling compressed-air tunnels presented a major problem, and most contractors experienced difficulty in maintaining temperatures at the shield face of less than 30°C.

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TABLE 5. Compressed-Air Details

Contract (1)	Section (2)	Number and type of low-pressure compressors (3)	Total output (m^3/min) (4)	Maximum pressure rating (bar) (5)	Maximum pressure used (6)	Installed capacity in terms of tunnel face area m^3/min/m^2 of face (7)	Man-lock size (8)	Material-lock size (9)	Total number of man decompressions (10)	Number of decompression sickness cases (11)	Incidence rate of decompression sickness (%) (12)
C104	Novena–Toa Payoh	Four-turbine blowers	200	1.5	1.3	4	Three compartments, 15 m × 1.5 m diameter	30.6 m × 2.8 m diameter	384,700	36 type 1, one type 2	0.05
	Novena–Newton	Four-turbine blowers	220	2.0	1.5	4.5	Three compartments, 15 m × 1.5 m diameter	30.5 m × 2.8 m diameter			
C105	Somerset–Orchard–Newton	Three-turbine blowers	113	2.5	1.0	4	9.14 m × 1.5 m diameter	30.48 m × 3.35 m diameter	145,000	Three type 1	0.01
C107	Raffles Place–City Hall	Two-turbine blowers	130	2.0	1.8	5	13 m × 1.6 m diameter	17 m × 2.5 m diameter	41,400	30 type 1	0.34
C108	Tanjong Pagar–Raffles Place	Two-turbine blowers	130	2.0	1.8	5	Three compartments, 16.4 m × 1.55 m diameter	22.5 m × 2.8 m diameter	87,600	25 type 1	0.14
C109	Outram Park–Tanjong Pagar	Four-turbine blowers	205	2.0	1.5	4	15 m × 1.5 m diameter	21 m × 2.7 m diameter	231,000	27 type 1, two type 2	0.08
C301	City Hall–Bugis	Two-turbine blowers, two reciprocating	228	4.0	2.35	4.5	Three compartments, 18.6 m × 1.8 m diameter	19.6 m × 3.2 m diameter	14,600	35 type 1	0.24

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TABLE 6. Segmental-Lining details

Contract (1)	Section (2)	Type/manufacturer (3)	Total length (mm) (4)	Number of segments/ring (5)	Internal diameter (mm) (6)	Thickness (mm) (7)	Bolt pockets (8)	Sealing strip (9)	Lining erector (type) (10)	Cavity grout (11)	Mixing location (12)	Placing method (13)
C104	Shan Road–Toa Payoh, Novena–Newton	Precast concrete/Hitashi Zosen	3,255	Five plus key	5,300	250	All steel	Butyl plus neoprene	Ring ± 200°	Cement/sand	Surface	Pipe to shield, monopumps on shield
C105	Orchard–Somerset (northbound)	Precast concrete/Hong Leong	840[a]	Six plus key	5,230	235	Steel, cross joint; concrete-circle joint	Neoprene	Ring ± 180°	Cement/sand	In tunnel	No. 2 pumps on shield
C106	City Hall–Dhoby Ghaut–Somerset	Precast concrete/Hong Leong	2,620[a]	Six plus key	5,230	235	Steel, cross joint; concrete-circle joint	Neoprene	Ring ± 180°	Cement/sand	Surface	1.5-m^3 tank on truck to shield; No. 2 pumps on shield
C107	Raffles Place–City Hall	Precast concrete/Hong Leong	2,490	Five plus key	5,400	225	All concrete	Butyl plus neoprene	Ring ± 200°	Bentonite/sand/cement	Surface	Agitator car to shield, piston action pumps on shield
C107B	City Hall–C301 interface	Precast concrete/Hong Leong	340	Five plus key	5,300	250	All concrete (curved bolts)	Hydrophilic rubber	Ring	Cement	Surface	3-m^3 agitator car fitted with monopump
C108	Tanjong Pagar–Raffles Place	Precast concrete/Hitachi Zosen	1,850	Five plus key	5,400	250	All steel	Composite butyl plus neoprene	Ring ± 220°	Cement/sand	Surface	3-m^3 agitator car fitted with monopump
C109	Tiong Bahru–Outram Park–Tanjong Pagar	Precast concrete/Hitachi Zosen	3,560	Five plus key	5,300	225	All concrete	Butyl	Ring	Cement/sand/bentonite	Surface	3-m^3-agitator car fitted with monopump
C301	End C107B–Bugis–Lavender	Precast concrete/Hong Leong	2,700	Five plus key	5,300	250	Steel, cross joint; concrete-circle joint with curved bolts	Hydrophilic rubber	Ring	Cement	Surface	3-m^3 agitator car to monopumps in tunnel

[a]All rings are tapers.

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PERMANENT LINING

Precast concrete segmental lining was used for 17 km of the 22.4 km of bored tunnel (Table 6). With the exception of C105 and C106, the contractors chose a ring of five segments plus key with thicknesses of 225–250 mm. All linings were manufactured in Singapore by two specialist precast manufacturers. A variety of bolt-pocket layouts was chosen. Steel bolt pockets were used by C104 and C108 on both segment faces. Bolt-pocket recesses formed in the concrete were used by C107, C107B, and C109. A combination of recessed pockets for the circle bolts and steel pockets on the cross joints were used by C105, C106, and C301. Curved bolts were used on C107B and C301 in conjunction with the recessed concrete bolt pockets. In all cases, bolt pockets were left unfilled.

The key segments chosen can all be described as large keys, but they varied in width from 500 mm to 1,060 mm (Fig. 3). With the exception of C105 and C106, all keys were rectangular, with the radial joint faces tapered to help insert the key. On C105 and C106, the key segment was tapered in plan in a similar manner to an expanded lining.

Tapered rings were employed on all contracts to negotiate curves, with a taper of 40 mm fairly standard. On C105 and C106 all rings were tapered.

To enhance corrosion protection of the segments, a two-part bitumen system, subsequently changed to coal-tar epoxy, was applied to the rear and side faces of all segments prior to delivery to site.

Segment erection was carried out using ring erectors in all cases. Segments were delivered to the face by flat bogies, then generally lifted and moved forward by chain block on a runway beam to be lowered in the tail skin invert. A positioning stirrup was fitted to the erector in all cases and could move longitudinally by 200–300 mm. The Grosvenor shields used on C105 and C106 had a segment magazine fitted below the trailing gantry, which delivered segments directly to the tail-skin invert.

In situ reinforced concrete lining was used for those lengths of tunnel not lined with precast segments. Between Somerset to Orchard and Orchard to Newton, a circular concrete lining cast in 22.5-m lengths was constructed at an average rate of 107 m/week. The remaining lengths of in situ lining were cast in two sections. Generally the invert was concreted first, followed

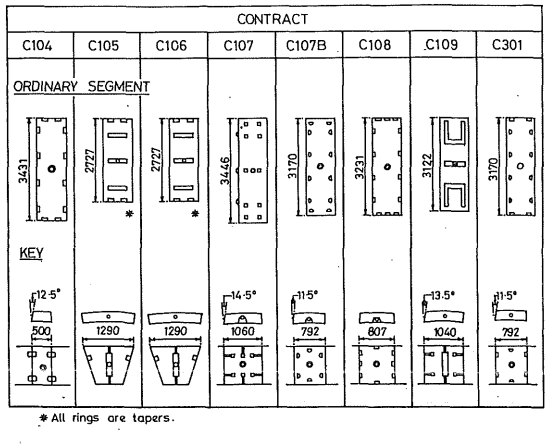


FIG. 3. Precast Concrete Segments

by the arch in 10.5-m lengths. On C107B, the contractor erected the arch first, followed by the invert and first-stage track concrete cast integrally.

WATER-SEALING TECHNIQUES

The corporation specified that the completed tunnels should achieve an overall watertightness of 5 ml/m²/hr or 10 ml/m²/hr over any 10-m length. This specification has been achieved but certain contractors have found difficulty in meeting the requirements.

Joint sealants were used throughout the segmentally lined tunnels. These varied from simple rectangular-section butyl rubber to water-swelling hydrophilic rubber sealing strips fixed to the segment trailing edge and one cross-joint face. Neoprene gaskets fitted into grooves around the complete outer edge of the segments were used on C105 and C106. The water-swelling hydrophilic rubber sealing strips produced significantly better results than all of the other systems.

On completion of lining construction, the general approach has been to carry out regrouting with cement grout and to use polyurethane grout at specific leakage points. Systematic caulking of the precast concrete (PCC) linings generally has not been done, allowing the corporation at a later stage the option of caulking as part of the ongoing tunnel maintenance.

For the in situ lined tunnels, the contractors attempted to achieve the watertightness specification before casting the in situ concrete lining. On C105, the temporary timber lagging even under compressed air proved inadequate to control water inflow. Between removal of the air pressure

TABLE 7. Settlements

Tunneling methods (1)	Kallang formation (marine clays/fluvial sands) (2)	Jurong formation (weathered to fresh sedimentaries) (mm) (3)	Bukit Timah formation (completely weathered/granite) (mm) (4)
Greathead shield	NA	10–200 (C106), 10–350 (C109)	10–45 (C105)[a], 10–60 (C106)
Greathead shield and compressed air	30–100 (C104), 20–60 (C301)	5–15 (C301), 100 (C109), 20 (C107)[b]	5–40 (C104)
Greathead shield and ground treatment	100–200 (C106)	NA	NA
Greathead shield with ground treatment and compressed air	30–60 (C108)	NA	NA
Tunnel-boring machine	NA	NA	50–300 (C105)[a]
Tunnel-boring machine with compressed air	200–400 (C105)	NA	20–120 (C105)
Earth-pressure balanced shield	60–120 (C301)	NA	NA
New Austrian tunneling method	NA	5–20 (C104), 5–10 (C107B), 20–60 (C107A)[c]	10–20 (C105)[a]
New Austrian tunneling method in compressed air	NA	NA	10–100 (C104)

[a]Single tunnel.
[b]Four tunnels.
[c]Twin tunnels, one vertically above the other.
Note: Settlements are for twin tunnels, side by side unless noted otherwise; NA = not applicable.

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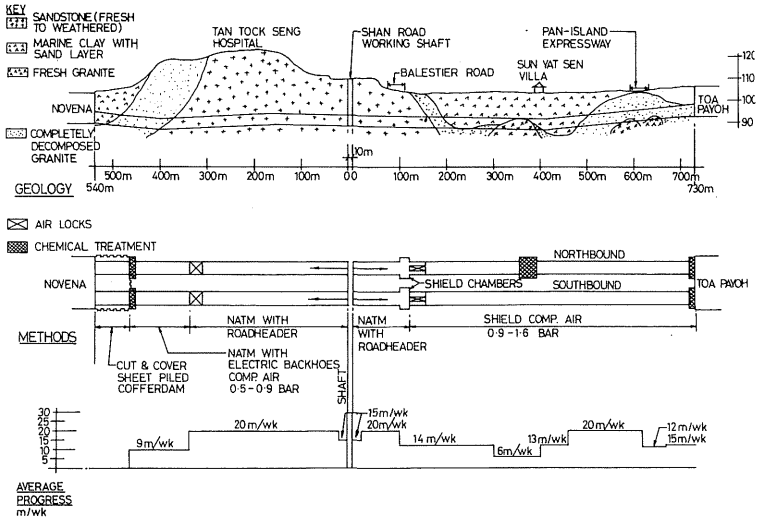


FIG. 4. Contract 104, Toa Payoh–Novena

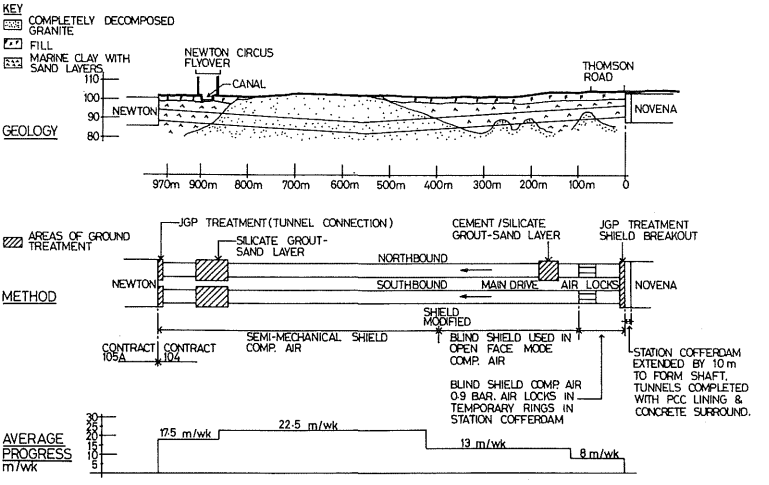


FIG. 5. Contract 104, Novena–Newton

and completion of the in situ lining in free air, there was considerable water ingress, and major (up to 350 mm) settlement occurred.

SETTLEMENTS

Ground-surface settlements above the bored tunnels varied considerably but can primarily be related to ground condition, construction methods, and workmanship (Table 7).

In the marine-clay sections, methods included Greathead shields with compressed air (two contracts), Greathead shield with jet-grouted–pile ground

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treatment and compressed air, TBM with compressed air, and earth-pressure balance shields. Of these methods, the Greathead shields with compressed air produced lower settlements than the full-face machines, but it was evident that air pressures approaching full hydrostatic pressure were required to achieve this. On one contract, where the tunnels crossed underneath a canal, air pressures inside the tunnel had to be reduced to minimize the risk of an air blow, but this in turn led to increased settlement to about 100 mm compared to 30–50 mm achieved earlier in the drive in similar conditions.

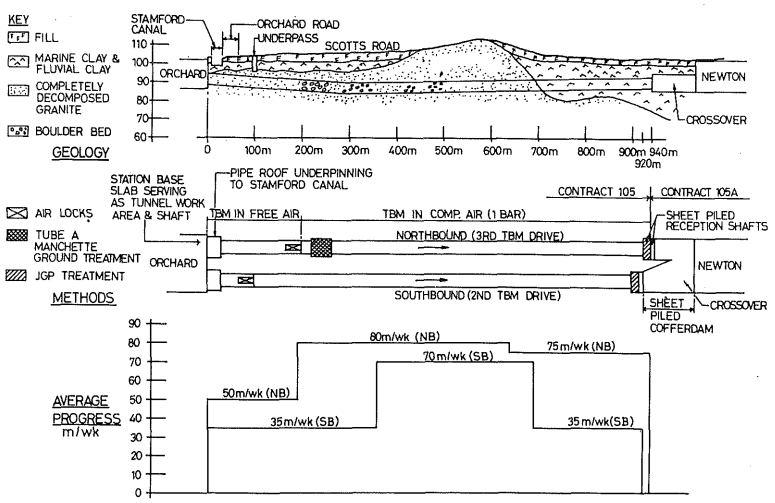


FIG. 6. Contract 105, Orchard–Newton

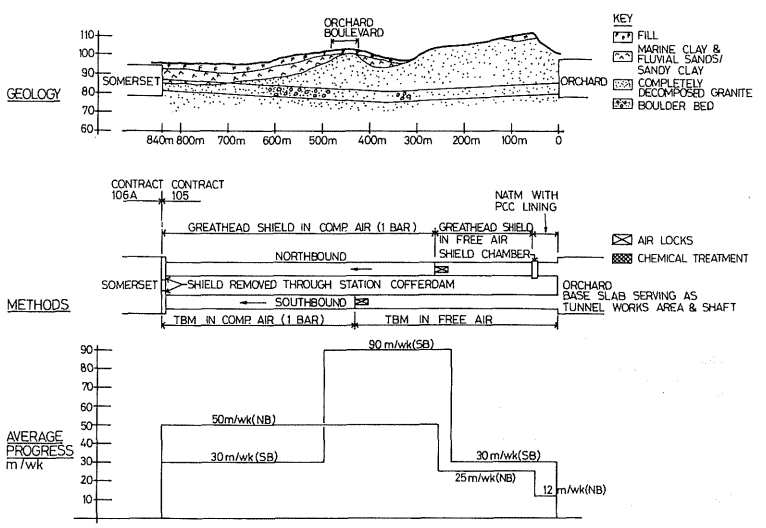


FIG. 7. Contract 105, Orchard–Somerset

The earth-pressure balance shields produced relatively small initial settlements of about 30 mm, but consolidation settlements over a period of six months after tunnel driving had ended increased the final values up to 120 mm. The TBM proved unsatisfactory with regard to settlements principally because there was no effective way to control the volume of soil removed as compared with the forward advance of the machine. Settlements on this contract reached 400 mm.

The sedimentary deposits and boulder clay layers were dealt with by Greathead shields in free air (two contracts), Greathead shields with compressed air (three contracts), and NATM (three contracts). Generally, the NATM produced lower surface settlements than the shield driven tunnels

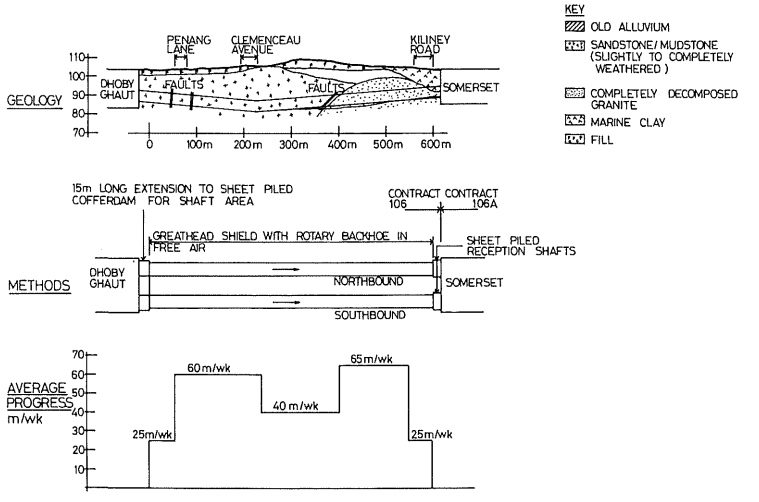


FIG. 8. Contract 106, Dhoby Ghout–Somerset

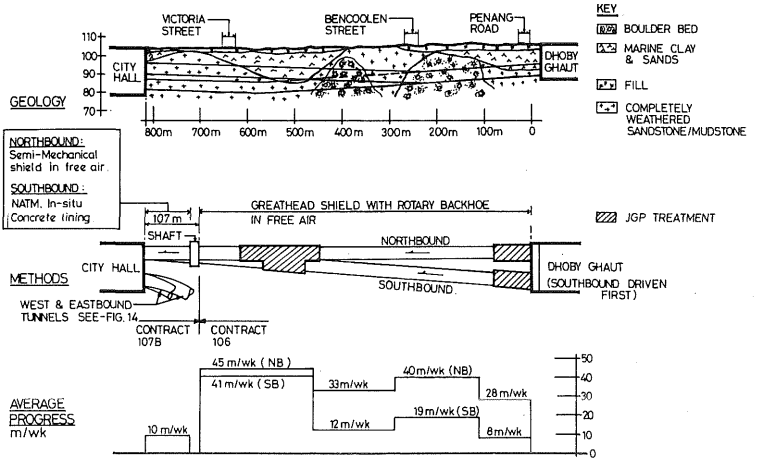


FIG. 9. Contract 106 and 107B, Dhoby Ghaut–City Hall

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with values up to 20 mm for twin, side-by-side tunnels, and 60 mm for two tunnels, one above the other. However, these NATM drives did not meet the friable, raveling conditions encountered in the completely weathered sedimentary deposits or the faulted zones found on other contracts; in these locations, large settlements of 350 mm and 200 mm, respectively, occurred due to difficulties in providing adequate face support within the open-face shields. Apart from these conditions, the shield drives generally produced settlements of between 10 mm and 50 mm in the sedimentaries and boulder clay.

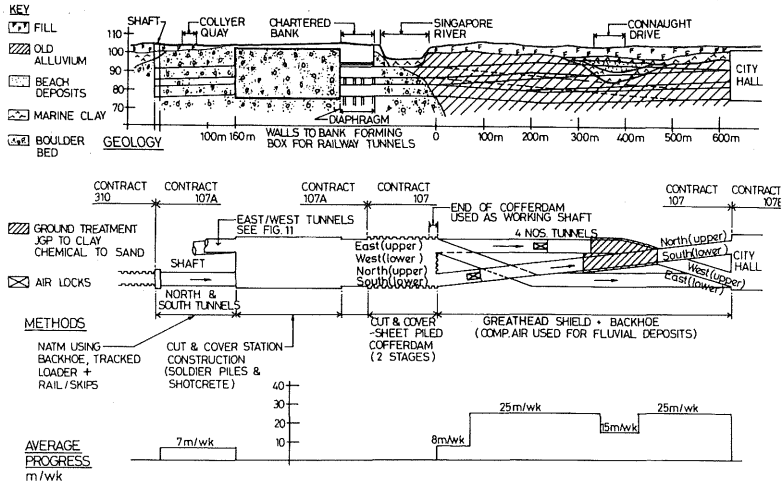


FIG. 10. Contract 107, Raffles Place–City Hall; Contract 107A, Raffles Place–C310 Interface

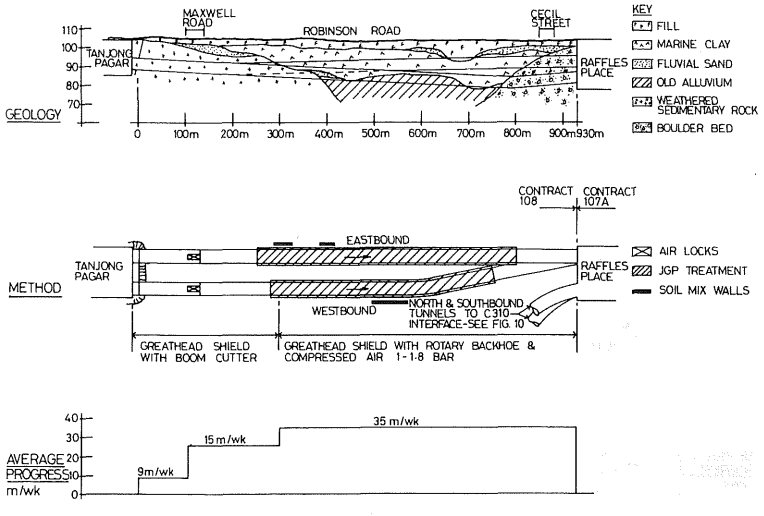


FIG. 11. Contract 108, Tanjong Pagar–Raffles Place

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Completely weathered granite was encountered in three contracts; Greathead shields in free air, Greathead shields in compressed air, a TBM in both free air and compressed air, and NATM methods in free and compressed air were used. The Greathead shields produced settlement in the range of 10–60 mm. The application of compressed air gave a slight reduction to this range. The TBM again proved unsatisfactory for the same reasons as with the marine-clay section, with the lack of control on volume of soil excavated giving settlements in free air up to 300 mm and in compressed air up to 120 mm.

The NATM solutions resulted in settlements up to 20 mm, except for a

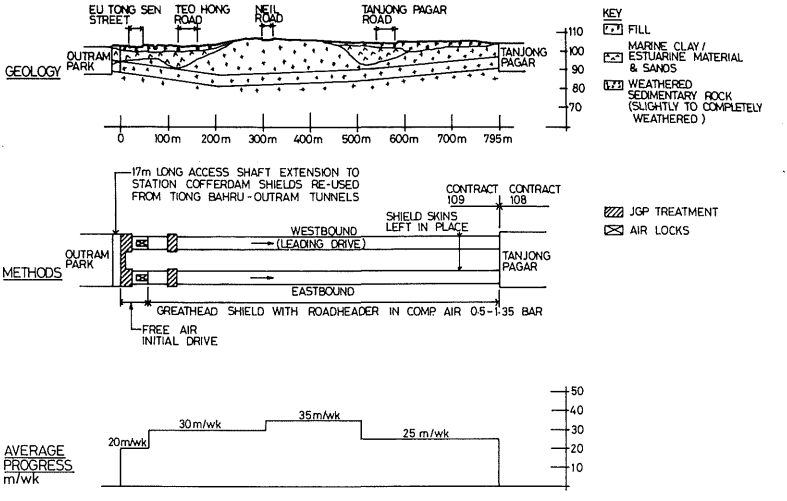


FIG. 12. Contract 109, Outram Park–Tanjong Pagar

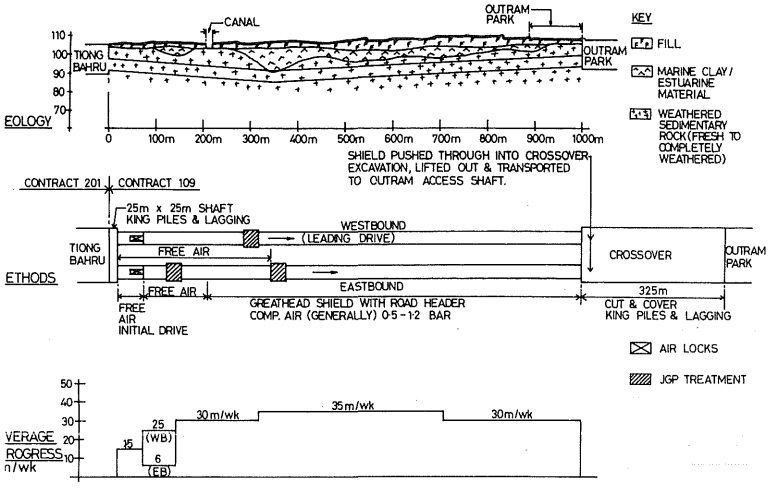


FIG. 13. Contract 109, Tiong Bahru–Outram Park

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length of C104 where problems were encountered with deformation of the second tunnel temporary lining and surface settlements reached 100 mm before the lining was reinforced and conditions stabilized.

In the segmental lined tunnel drives, the close supervision provided by the corporation ensured that rings were almost invariably grouted immediately behind the tail skin and the subsequent back grouting carried out at regular interests. This is considered to have assisted in restricting settlement due to ground loss.

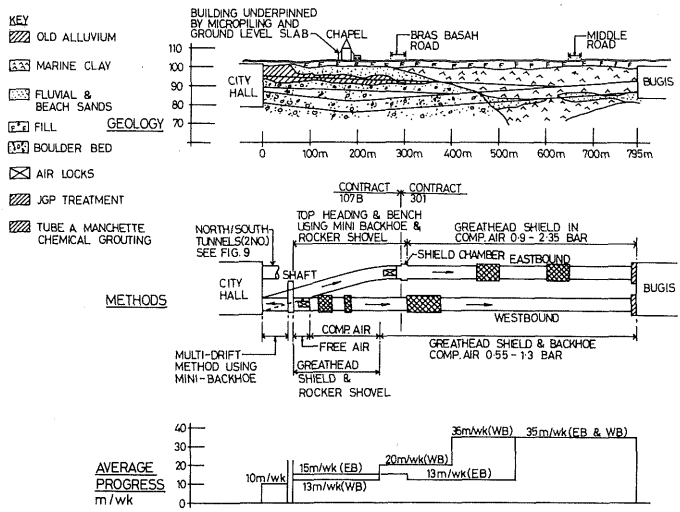


FIG. 14. Contract 107B and 310, City Hall–Bugis

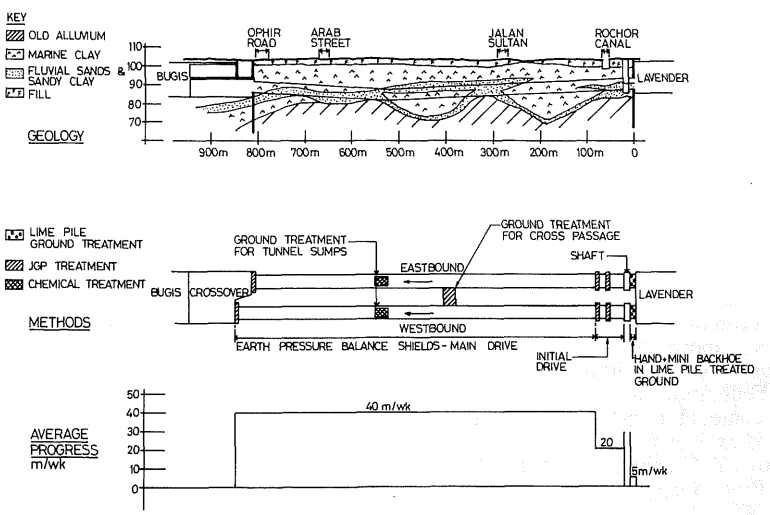


FIG. 15. Contract 301, Bugis–Lavender

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TUNNELING RATES

The tunneling methods adopted and the average advance rates achieved in each drive are shown in Figs. 4–15. In addition, the geological conditions and the various ground treatments are shown as having a direct relationship to the tunneling rates attained in the corresponding length of drive.

In shield drives, rates for a three-shift, six-day week varied from 6 m up to 65 m almost entirely dictated by the ground conditions.

TBM rates up to 80 m/week were attained but were often accompanied by large settlements. EPBM rates averaging 40 m/week were very consistently attained.

NATM drive rates varied between 9 m/week in compressed air up to 20 m/week in free air.

This very great disparity in tunneling advance rates attained in superficially similar drives emphasizes the significant effects relatively minor changes in geology can have on progress.

CONCLUSIONS

In most tunneling, the safe and timely completion of the drives within budget is the principal objective.

From this account of the Singapore Metro tunneling, some general conclusions can be deduced that will help in the future:

1. For contracts with more than two drives, a minimum of two shields or TBM are required to guard against significant delays in any one drive impacting on the whole program.
2. In the short drives (less than 1 km) usually involved in rapid-transit tunnels, relatively simple Greathead-type shields with mechanized excavation are generally preferred to more sophisticated tunnel-boring machines.
3. In urban areas soft-ground tunneling under hydrostatic head must be accompanied by the immediate erection of a permanent or semipermanent lining, which can be effectively sealed to minimize water inflow and reduce consolidation settlement.
4. Compressed air, even when its use is only precautionary, has as yet no effective substitute in minimizing settlement. Closed-face shields are being steadily improved, however, and may soon offer a viable alternative.
5. Hydrophilic sealing strips between the precast segments are one of the most effective means of minimizing water inflow through a segmental tunnel lining.

APPENDIX. REFERENCES

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